

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1576

Commonwealth Chesapeake 138 kV
January 23, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1576. The Interconnection Customer proposes a 149.5 MW solar photovoltaic facility interconnecting at the Commonwealth Chesapeake 138 kV substation in the Virginia area. The request was received on September 25, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 526 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,083,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N Lake – Kiwi 138 KV	Easton 2 – Vienna Operations 345 KV line	127.6	528	24.5
Sandburg – Jarratt 138 KV	Vienna Operations – Easton 2 345 KV	127.4	542	27.0
Sherwood – Sandburg 138 KV	Vienna Operatons – Easton 2 Substation 345 KV L	126.7	554	27.0
Hitchland – Sherwood 138 KV line	Vienna Operations – Easton 2 345 KV	126.3	561	27.0
North Lake – Hitchland 138 KV	Vienna Operations – Easton 2 Substation 345 KV	124.5	591	27.0
Chemolite – Lammers 138 KV	Easton 2 – Vienna Operations 345 KV	123.8	318	15.3
Ess Lewes – Newmrk 138 KV	Easton 2 – Vienna Operations 345 KV	122.2	427	25.1
St Bonifacius – West -Station 138 KV	Vienna perations – Southtown 345 KV	116.2	1,362	53.6
Wye Mills – Holston Dam 138 KV	Easton 2 – Cady 345 KV line	116.0	231	6.4
Chemolite – Randall County 138 KV	Vienna Operations – Eason 2 345 KV Line	115.2	330	15.3
Park Ave – Hillsboro 138 KV	Cady – Easton 2 345 KV	112.9	760	24.8
Carancahua – Holston Dam 138 KV	Easton 2 – Cady 345 KV	109.4	235	6.0
Park Ave – Harbor 138 KV	Cady – Easton 2 345 KV L	108.8	410	12.2
Cady 345/138 kV Transformer Bank	Cady – Dodd Rd 345 KV L	108.7	823	26.7
Robinsonville – Angola 138 KV Line	Quinn Substation – Bulverde 138 KV Line	108.1	180	6.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Harbor Substation – Holston Dam 138 KV line	Cady – Easto 2 345 KV	108.1	432	12.2
Grasonville – Macon 138 KV	Cady – Easton 2 345 KV line	107.6	433	12.4
Oak Hill – Louisa Substation 138 KV	Dodd Rd – Cady 345 KV	107.1	324	11.0
Mesquite – Arroyo Tap 138 KV	Sothtown – Vienna Operations Substation 345 KV	106.7	1,151	46.2
Cady – Louisa 138 KV L	Cady – Dodd Road 345 KV line	106.6	338	11.0
Ocean City – Reidland 138 KV	Southtown Substation – Vienna Operations 345 KV L	106.4	1,171	46.2
Reidland – Arroyo Tap 138 KV	Southtown Substation – Vienna Operations 345 KV Line	106.4	1,171	46.2
Candelaria Arenas 1718 – Ocean City 138 KV	Southtown – Vienna Operations 345 KV	106.2	1,187	46.2
Lost Hills – Oak Hill 138 KV	Cady – Dodd Rd 345 KV	105.6	466	15.9
Cady – KeyLargo 138 KV	Cady – Dodd Rd Substation 345 KV	103.4	214	6.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Lake – Kiwi 138 KV	Rebuild line with higher-capacity conductor	\$5,351,000	5% (\$269,000)
Sandburg – Jarratt 138 KV	Rebuild line with higher-capacity conductor	\$7,110,000	5% (\$356,000)
Sherwood – Sandurg 138 KV	Rebuild line with higher-capacity conductor	\$3,974,000	5% (\$199,000)
Hitchland – Sherwood 138 KV line	Rebuild line with higher-capacity conductor	\$4,967,000	5% (\$249,000)
Hitchland – N Lake Substation 138 KV line	Rebuild line with higher-capacity conductor	\$6,909,000	5% (\$346,000)
Chemolite – Lammers 138 KV	Rebuild line with higher-capacity conductor	\$4,470,000	5% (\$222,000)
Ess Lewes – Newmark 138 KV	Rebuild line with higher-capacity conductor	\$9,470,000	5% (\$482,000)
West -Station – St Bonifacius 138 KV	Rebuild line with higher-capacity conductor	\$6,331,000	5% (\$301,000)
Wye Mills – Holston Dam 138 KV	Rebuild line with higher-capacity conductor	\$8,753,000	4% (\$373,000)
Randall County – Chemolite 138 KV L	Rebuild line with higher-capacity conductor	\$7,119,000	5% (\$354,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Park Ave – Hillsboro 138 KV	Rebuild line with higher-capacity conductor	\$5,686,000	4% (\$247,000)
Carancahua – Holston Dam 138 KV L	Reconductor line	\$3,530,000	4% (\$150,000)
Harbor – Park Ave 138 KV L	Reconductor line	\$5,628,000	4% (\$230,000)
Cady 345/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$21,593,000	4% (\$970,000)
Robinsonville – Angola 138 KV Line	Reconductor line	\$2,311,000	5% (\$115,000)
Harbor Substation – Holston Dam 138 KV line	Reconductor line	\$7,287,000	4% (\$298,000)
Grasonville – Macon 138 KV	Reconductor line	\$627,000	4% (\$26,000)
Oak Hill – Louisa Substation 138 KV	Reconductor line	\$1,188,000	5% (\$55,000)
Mesquite – Arroyo Tap 138 KV	Reconductor line	\$1,299,000	5% (\$70,000)
Cady Substation – Louisa Substation 138 KV	Reconductor line	\$1,226,000	5% (\$56,000)
Ocean City – Reidland 138 KV	Reconductor line	\$1,154,000	5% (\$62,000)
Reidland – Arroyo Tap 138 KV	Reconductor line	\$700,000	5% (\$38,000)
Candelaria Arenas 1718 – Ocean City 138 KV	Reconductor line	\$1,539,000	5% (\$83,000)
Lost Hills – Oak Hill 138 KV	Reconductor line	\$8,862,000	4% (\$389,000)
Cady – Key Largo 138 KV	Reconductor line	\$2,806,000	5% (\$143,000)
	Total allocated to project		\$6,083,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.